

Revised Generation Interconnection System Impact Study Report

For Queue Project Q12-0119

Monserate 138 kV
April 21, 2013

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q12-0119. The Interconnection Customer proposes a 327.6 MW battery energy storage facility interconnecting at the Monserate 138 kV substation in the New Jersey area. The request was received on March 24, 2012. The study evaluated system intact and single contingency conditions using the regional planning case, including 101 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 11 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$25,782,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Monserate – Parlin Power Plant 138 KV L
Contingency: Sarepta – Read Oak Station 138 KV
Loading: 206.8% (Rating: 195 MVA)
DFAX: 100.0%
Pre-project loading: 39.2%
2. Facility: Sarepta – Read Oak Sta 138 KV L
Contingency: Monserate – Parlin Power Plant 138 KV
Loading: 166.5% (Rating: 243 MVA)
DFAX: 100.0%
Pre-project loading: 31.5%
3. Facility: Monerate – Sarepta 138 KV
Contingency: Monserate – Parlin Power Plant 138 KV
Loading: 164.2% (Rating: 234 MVA)
DFAX: 100.0%
Pre-project loading: 23.9%
4. Facility: Rose – Parlin Power Plant 138 KV
Contingency: Sarepta – Read Oak Sta 138 KV
Loading: 159.0% (Rating: 166 MVA)
DFAX: 55.8%
Pre-project loading: 49.2%
5. Facility: Mill C – Rose 138 KV
Contingency: Read Oak Sta Substation – Sarepta 138 KV
Loading: 134.6% (Rating: 242 MVA)
DFAX: 47.8%
Pre-project loading: 69.9%

6. Facility: Sayreville Cogeneration Facility – Read Oak Sta 138 KV
Contingency: Monserate Substation – Parlin Power Plant 138 KV L
Loading: 118.6% (Rating: 236 MVA)
DFAX: 56.7%
Pre-project loading: 40.0%
7. Facility: Sayreville Cogeneration Facility – Mill C 138 KV
Contingency: Read Oaksta 345 kV/138 kV Transformer Bank
Loading: 112.1% (Rating: 217 MVA)
DFAX: 45.2%
Pre-project loading: 43.7%
8. Facility: Devine – Caparra 1924 138 KV
Contingency: Mill C – Rose 138 KV
Loading: 102.5% (Rating: 262 MVA)
DFAX: 14.2%
Pre-project loading: 84.8%
9. Facility: Wallenpaupack 500/345 kV XFMR
Contingency: W Gallatin – Walnut 345 KV
Loading: 102.0% (Rating: 1,152 MVA)
DFAX: 11.2%
Pre-project loading: 98.8%
10. Facility: Curecanti – Eliza Creek 345 KV
Contingency: Tombstone – Bixby Hill 500 KV
Loading: 101.6% (Rating: 1,135 MVA)
DFAX: 14.5%
Pre-project loading: 97.4%
11. Facility: Talenenergy Martins Crk Llc Harwood Substation 345 kV/1 kV TX
Contingency: Contadina – Warm Springs 345 KV
Loading: 100.1% (Rating: 1,053 MVA)
DFAX: 6.3%
Pre-project loading: 98.2%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Monserate – Parlin Power Plant 138 KV L	Rebuild line with higher-capacity conductor	\$2,236,000	100% (\$2,236,000)
Sarepta – Read Oak Station 138 KV line	Rebuild line with higher-capacity conductor	\$1,681,000	100% (\$1,681,000)
Monerate – Sarepta 138 KV	Rebuild line with higher-capacity conductor	\$1,110,000	100% (\$1,110,000)
Rose – Parlin Power Plant 138 KV	Rebuild line with higher-capacity conductor	\$7,707,000	100% (\$7,707,000)
Mill C – Rose 138 KV	Rebuild line with higher-capacity conductor	\$1,537,000	100% (\$1,537,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Sayreville Cogeneration Facility – Read Oak Sta 138 KV	Rebuild line with higher-capacity conductor	\$2,010,000	100% (\$2,010,000)
Mill C – Sayreville Cogeneration Facility 138 KV	Rebuild line with higher-capacity conductor	\$1,094,000	100% (\$1,094,000)
Devine – Caparra 1924 Substation 138 KV Line	Reconductor line	\$185,000	100% (\$185,000)
Wallenpaupack 500/345 kV Transformer Bank	Replace transformer with higher-rated unit	\$35,371,000	15% (\$5,326,000)
Curecanti – Eliza Creek 345 KV	Reconductor line	\$7,644,000	12% (\$904,000)
Talenenergy Martins Crk Llc Harwood 345/1 kV TX	Replace transformer with higher-rated unit	\$15,729,000	13% (\$1,992,000)
	Total allocated to project		\$25,782,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.